

# Relation Algebra

## §15. Dynamic Programming

### §15.1. Theory

$h : FB \rightarrow B$  a map,  $T : FA \rightarrow A$  an  $F$ -algebra,  $R : B \rightarrow B$ .

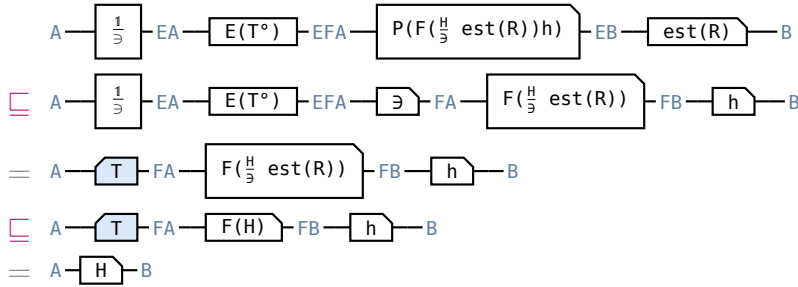
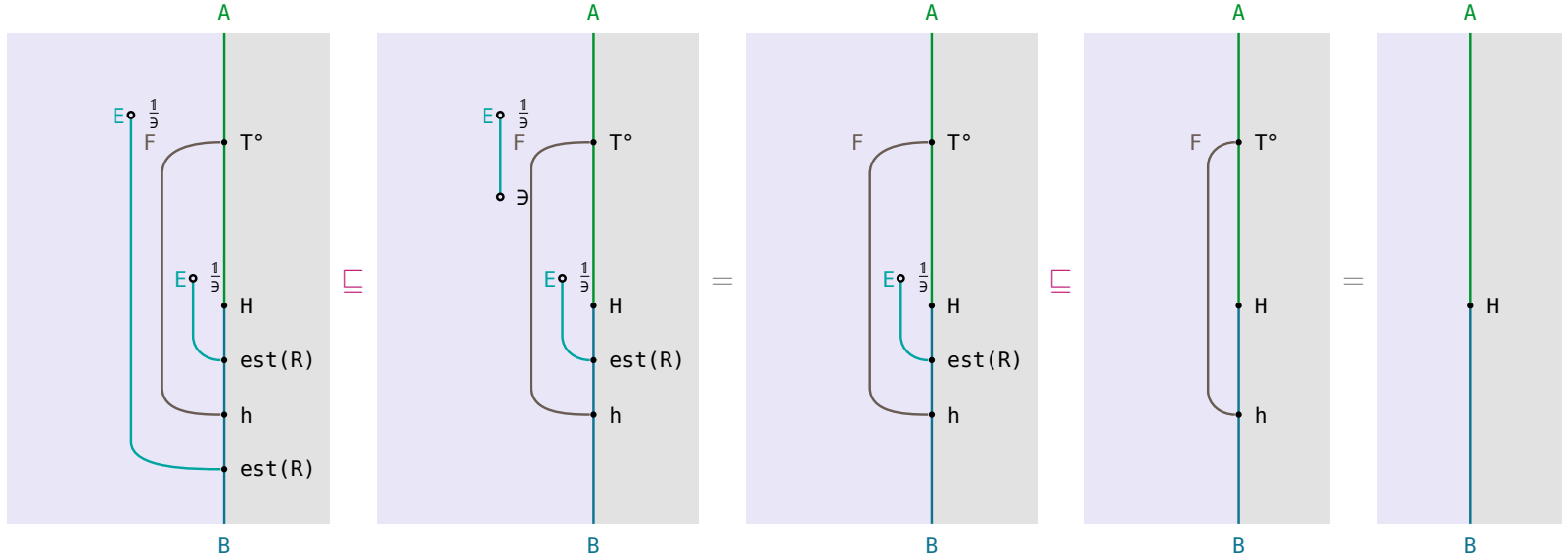
(15.1a)

$H \triangleq (T)^\circ \circ (h) : A \rightarrow B$ ,  $M \triangleq \frac{H}{\exists} \text{ est}(R)$  the problem to be solved,  $(\mu X : G(X))$  as in  $\triangleq \mu$ .

$$T^\circ F(H)h = H \Rightarrow \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{ est}(R))h) \text{ est}(R) \sqsubseteq H$$

(15.1b)

taking the input apart every way  $T$  allows ( $\frac{T^\circ}{\exists}$ ), solving each part by  $M$  and keeping an optimum ( $P(F(M)h) \text{ est}(R)$ ) returns only what  $H$  returns



$$(9.4) P(X) \text{ est}(R) \sqsubseteq \exists X \text{ at } X \triangleq F(\frac{H}{\exists} \text{ est}(R))h \text{ — est-710}$$

$$\frac{T^\circ}{\exists} \exists = T^\circ \text{ — pow-laws}$$

$$M \triangleq \frac{H}{\exists} \text{ est}(R) \sqsubseteq \frac{H}{\exists} \exists = H \text{ — est-up}$$

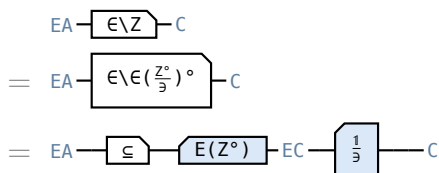
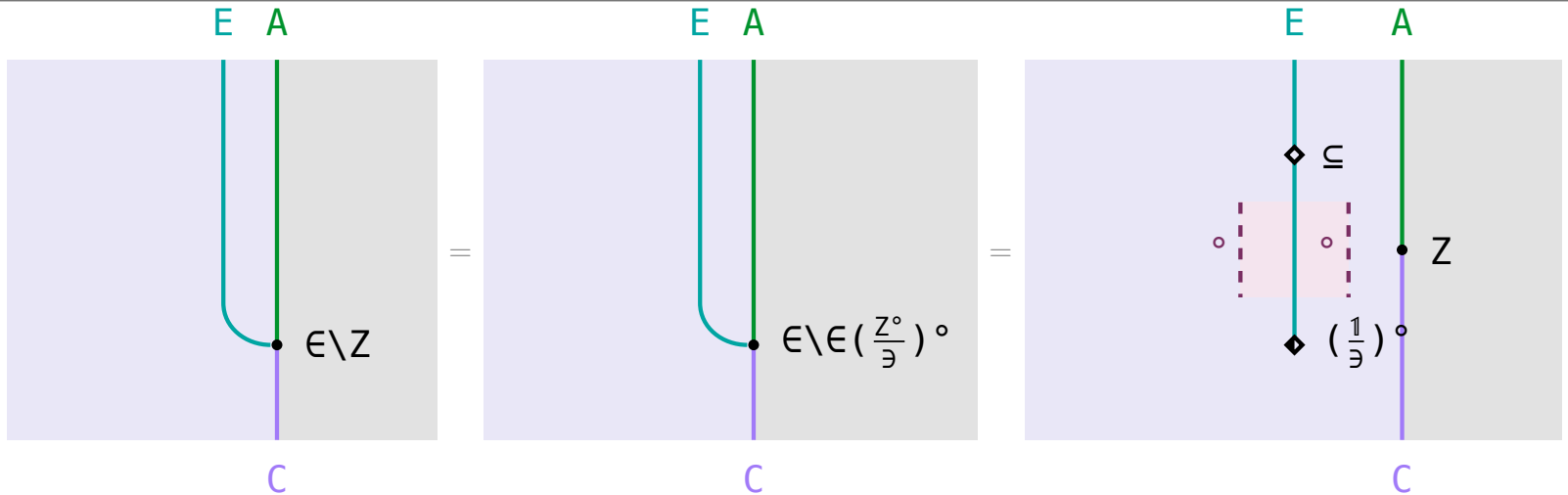
$$T^\circ F(H)h = H: H \triangleq (T)^\circ \circ (h) \text{ and hylo-fix}$$

#### §15.1.1. $\epsilon \setminus$ as a composite

$$\epsilon \setminus Z = \epsilon \setminus (\frac{Z^\circ}{\exists})^\circ$$

(15.1.1a)

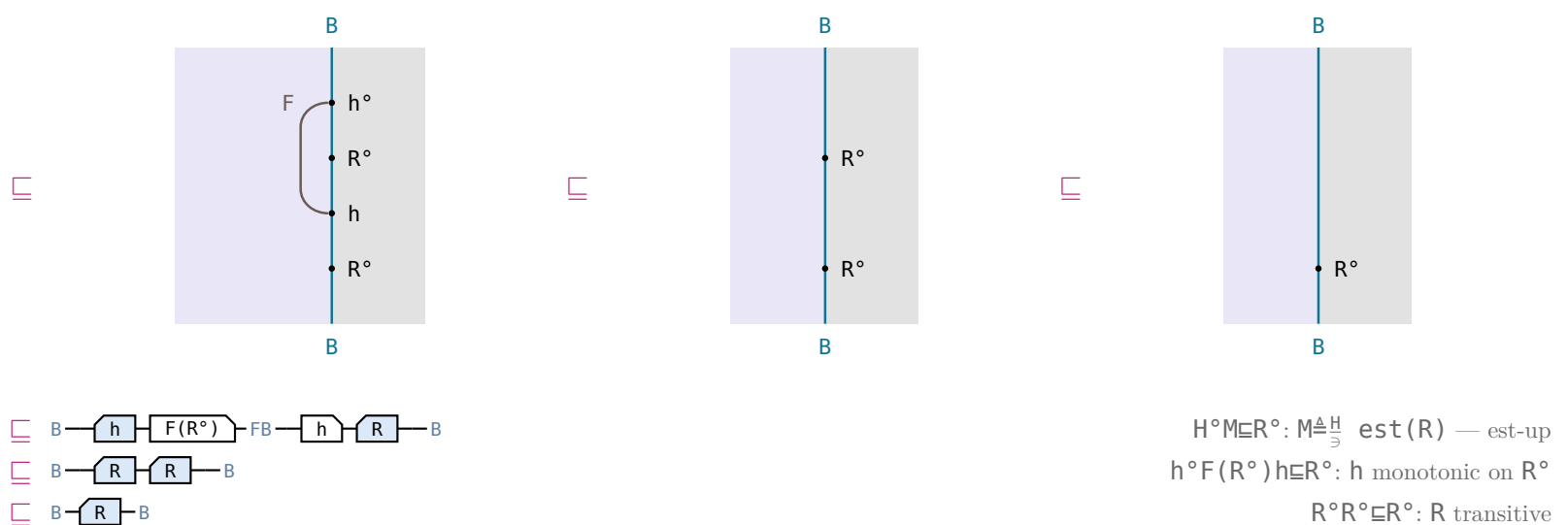
$xs$  is related to  $c$  by  $\epsilon \setminus Z$  exactly when  $xs \sqsubseteq \Lambda(Z^\circ)(c)$ , the set of every  $a$  with  $a Z c$



$$Z = \epsilon \setminus \Lambda(Z^\circ)^\circ: \text{the converse of } \Lambda(Z^\circ) \exists = Z^\circ \text{ — pow-laws}$$

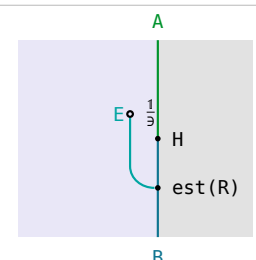
$$X \setminus (Y f^\circ) = (X \setminus Y) f^\circ \text{ for a map } f \text{ (not a tabulated row), at the map } f \triangleq \Lambda(Z^\circ); \epsilon \setminus \triangleq \epsilon \setminus \epsilon$$

for every  $\mathbf{b}$  that  $\mathbf{H}$  returns from an input, the step  $\frac{T_0}{\Xi} \mathbf{P}(\mathbf{F}(\mathbf{M})\mathbf{h}) \mathbf{est}(\mathbf{R})$  returns from that input only  $\mathbf{b}'$  with  $\mathbf{R}$  relating  $\mathbf{b}'$  to  $\mathbf{b}$



an optimum over everything  $\mathbf{H}$  returns is reached by taking the input apart every way  $\mathbf{T}$  allows, dropping the parts that can never win, solving each of the rest and keeping one optimum

Hinze–Marsden



| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Hinze–Marsden |
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| <p> <math>A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} P(F(X)h) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B</math> </p> <p> <math>\sqsubseteq</math> <math>h</math> monotonic on <math>R</math> and <math>Q</math> a preorder with <math>QF(H)h \sqsubseteq F(H)hR</math>; <math>\text{thin}(Q)</math> as in thin-laws. Theorem 9.1 is this with the thinning step dropped — <math>1 \sqsubseteq \text{thin}(Q)</math>. Knaster–Tarski leaves (9.1): the body at <math>M</math> is <math>\sqsubseteq M</math>; <math>M = Hn(H^\circ \setminus R^\circ)</math> splits that into (9.2) and (9.3) below — est-up. The fixed point is unique and entire when <math>T^\circ</math> followed by <math>F</math>'s membership relation is inductive, <math>\frac{T^\circ}{\exists}</math> finite and non-empty, <math>R</math> connected                 </p> |               |
| <p> <math>A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} P(F(\frac{H}{\exists} \text{est}(R))h) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B</math> </p> <p>(9.2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <p> <math>A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} \exists \xrightarrow{FA} F(\frac{H}{\exists} \text{est}(R)) \xrightarrow{FB} h \xrightarrow{} B</math> </p> <p>(9.4) <math>P(X)\text{est}(R) \sqsubseteq \exists X</math> at <math>X \triangleq F(\frac{H}{\exists} \text{est}(R))h</math> — est-710</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <p> <math>A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\exists} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B</math> </p> <p><math>\text{thin}(Q) \exists \exists \exists</math> — thin-laws</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <p> <math>A \xrightarrow{T} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B</math> </p> <p><math>\frac{T^\circ}{\exists} \exists = T^\circ</math> — pow-laws</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |

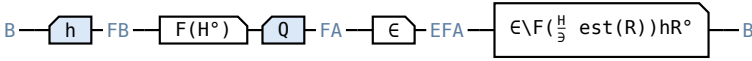
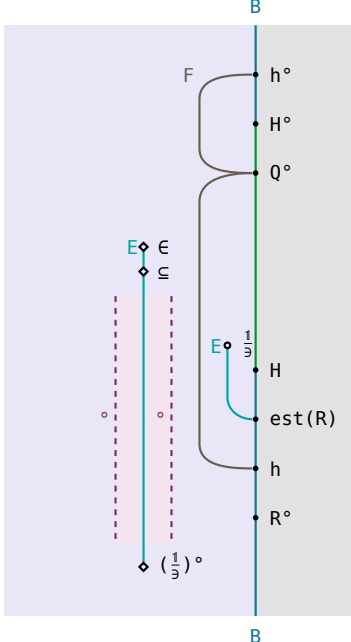
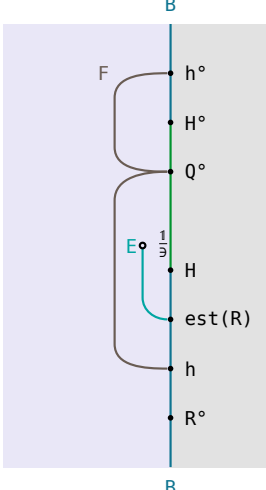
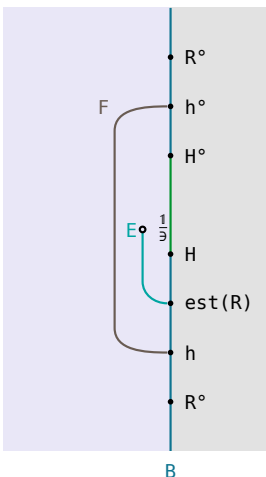
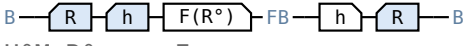
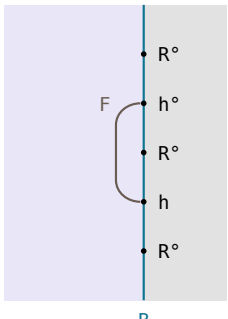
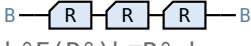
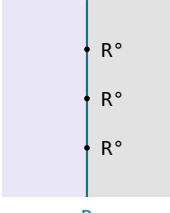
(15.1.1d)

| circuit                                                                                                                                                                                                                                                                                     | Hinze–Marsden |
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| $\begin{array}{c} \text{A} \text{---} \boxed{\text{T}} \text{---} \text{FA} \text{---} \boxed{\text{F(H)}} \text{---} \text{FB} \text{---} \boxed{\text{h}} \text{---} \text{B} \\ \text{M} \sqsubseteq \frac{\text{H}}{\text{F}} \text{---} \text{H} \text{---} \text{est-up} \end{array}$ |               |
| $\begin{array}{c} \text{A} \text{---} \boxed{\text{H}} \text{---} \text{B} \\ \text{T}^\circ \text{F(H)} \text{h} = \text{H}: \text{H} \triangleq (\text{T})^\circ \circ (\text{h}) \text{ and hylo-fix} \end{array}$                                                                       |               |

(15.1.1e)

| $\text{PreservesRecip(F)} \wedge \text{Map(h)} \wedge \text{MonotonicAlg(h, R}^\circ) \wedge \text{R}^\circ \text{R}^\circ \sqsubseteq \text{R}^\circ \wedge \text{T}^\circ \text{F(H)} \text{h} = \text{H} \wedge \text{QF(H)} \text{h} \sqsubseteq \text{F(H)} \text{hR} \implies \text{H}^\circ \frac{\text{T}^\circ}{\text{F}} \text{thin(Q)} \text{P(F(H)} \frac{\text{H}}{\text{F}} \text{est(R))h)est(R)} \sqsubseteq \text{R}^\circ$                                                                                                                                            |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\text{H}^\circ$ followed by the body at $\text{M}$ is $\text{ER}^\circ$ : an answer of the body is never worse than an answer of $\text{H}$ to the same input                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden |
| $\text{B} \text{---} \boxed{\text{H}} \text{---} \text{A} \text{---} \boxed{\frac{1}{\text{F}}} \text{---} \text{EA} \text{---} \boxed{\text{E(T}^\circ\text{)}} \text{---} \boxed{\text{thin(Q)}} \text{---} \text{EFA} \text{---} \boxed{\text{P(F(H)} \frac{\text{H}}{\text{F}} \text{est(R))h)}} \text{---} \text{EB} \text{---} \boxed{\text{est(R)}} \text{---} \text{B}$                                                                                                                                                                                                         |               |
| $\begin{array}{c} \text{B} \text{---} \boxed{\text{H}} \text{---} \text{A} \text{---} \boxed{\frac{1}{\text{F}}} \text{---} \text{EA} \text{---} \boxed{\text{E(T}^\circ\text{)}} \text{---} \boxed{\text{thin(Q)}} \text{---} \text{EFA} \text{---} \boxed{\text{E} \setminus \text{F(H)} \frac{\text{H}}{\text{F}} \text{est(R))hR}^\circ} \text{---} \text{B} \\ (9.4) \text{P(X)} \text{est(R)} \sqsubseteq \text{E} \setminus \text{X} \text{R}^\circ \text{ at } \text{X} \triangleq \text{F(H)} \frac{\text{H}}{\text{F}} \text{est(R))h} \text{---} \text{est-710} \end{array}$ |               |



| circuit                                                                                                                                                                                                                                  | Hinze–Marsden                                                                         |
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|                                                                                                                                                         |    |
|  <p><math>E(E \setminus Y) \sqsubseteq Y</math>, not a tabulated row: division cancels</p>                                                            |   |
|  <p><math>QF(H) h \sqsubseteq F(H) h R</math> conversed — the hypothesis on <math>Q</math></p>                                                        |  |
|  <p><math>H^\circ M \sqsubseteq R^\circ</math> under <math>F</math> — est-up</p>                                                                      |  |
|  <p><math>h^\circ F(R^\circ) h \sqsubseteq R^\circ</math>: <math>h</math> monotonic on <math>R</math>, shunted — the hypothesis on <math>h</math></p> |  |

| circuit                                                                                                          | Hinze–Marsden |
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| $\begin{array}{c} B \text{---} \boxed{R} \text{---} B \\ \sqsubseteq \\ R \text{ transitive, twice} \end{array}$ |               |

$$V_2 \quad V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) P([U_1, U_2]) \text{est}(R) = \text{ran}(V_1) \frac{V_1^\circ}{\exists} \text{thin}(Q_1) P(U_1) \text{est}(R) \cup \text{ran}(V_2) \frac{V_2^\circ}{\exists} \text{thin}(Q_2) P(U_2) \text{est}(R)$$

(15.1.1g)

when  $V_1$  and  $V_2$  have disjoint ranges, the body over  $[V_1, V_2]$  runs the  $V_1$  problem on inputs  $V_1$  reaches and the  $V_2$  problem on inputs  $V_2$  reaches

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Hinze–Marsden |
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| $A \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E([V_1, V_2]^\circ) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| $\begin{array}{c} A \text{---} \text{ran}(V_1) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E([V_1, V_2]^\circ) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \\ \cup \\ A \text{---} \text{ran}(V_2) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E([V_1, V_2]^\circ) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \end{array}$ <p>off <math>\text{ran } V_1 \cup \text{ran } V_2</math> the set is empty and <math>\text{est}(R)</math> of it is nothing; not a tabulated row</p>                                                                                                    |               |
| $\begin{array}{c} A \text{---} \text{ran}(V_1) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_1^\circ) \text{---} E\alpha \text{---} P(l) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \\ \cup \\ A \text{---} \text{ran}(V_2) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_2^\circ) \text{---} E\beta \text{---} P(r) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \end{array}$ <p>Exercise 9.5: on <math>\text{ran } V_1</math>, <math>\frac{[V_1, V_2]^\circ}{\exists} = \frac{V_1^\circ}{\exists} P(\text{in } l)</math>, as <math>V_1, V_2</math> have disjoint ranges</p> |               |
| $\begin{array}{c} A \text{---} \text{ran}(V_1) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_1^\circ) \text{---} \text{thin}(Q_1) \text{---} E\alpha \text{---} P(l) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \\ \cup \\ A \text{---} \text{ran}(V_2) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_2^\circ) \text{---} \text{thin}(Q_2) \text{---} E\beta \text{---} P(r) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \end{array}$ <p>Exercise 9.5: <math>P(\text{in } l) \text{thin}(Q_1 + Q_2) = \text{thin}(Q_1) P(\text{in } l)</math></p>                                                                                                |               |

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Hinze–Marsden |
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| $  \begin{aligned}  & A \rightarrow \boxed{\text{ran}(V_1)} \rightarrow \boxed{\frac{1}{\beta}} \rightarrow EA \rightarrow \boxed{E(V_1^\circ)} \rightarrow \boxed{\text{thin}(Q_1)} \rightarrow E\alpha \rightarrow \boxed{P(U_1)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B \\  = & A \rightarrow \boxed{\text{ran}(V_2)} \rightarrow \boxed{\frac{1}{\beta}} \rightarrow EA \rightarrow \boxed{E(V_2^\circ)} \rightarrow \boxed{\text{thin}(Q_2)} \rightarrow E\beta \rightarrow \boxed{P(U_2)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B \\  & P \text{ a relator and } \text{inl}[U_1, U_2] = U_1  \end{aligned}  $ |               |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |
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| $  \text{Map}(\text{cost}) \wedge R = \text{cost} \text{ leq } \text{cost}^\circ \wedge h \text{cost} = F(\text{cost}) k \wedge F(\text{leq}) k \leq k \text{leq} \implies \text{MonotonicAlg}(h, R)  $ <p>if <math>R</math> compares two values by comparing their <b>costs</b> under <math>\leq</math>, and <math>h</math> then <b>cost</b> equals <math>F(\text{cost})</math> then a <math>k</math> monotonic on <math>\leq</math>, then <math>h</math> is monotonic on <math>R</math></p> | (15.1.1h) |
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| circuit                                                                                                                                                                                                                                                                                           | Hinze–Marsden |
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| $  \begin{aligned}  & FA \rightarrow \boxed{F(R)} \rightarrow \boxed{h} \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow C \\  & F(R) h \sqsubseteq h R \text{ iff } F(R) h \text{ cost} \sqsubseteq h \text{ cost} \leq: \text{definition of } R \text{ and shunting}  \end{aligned}  $ |               |
| $  \begin{aligned}  = & FA \rightarrow \boxed{F(R)} \rightarrow \boxed{F(\text{cost})} \rightarrow FC \rightarrow \boxed{k} \rightarrow C \\  & \text{assumption } h \text{ cost} = F(\text{cost}) k  \end{aligned}  $                                                                            |               |
| $  \begin{aligned}  \sqsubseteq & FA \rightarrow \boxed{F(\text{cost})} \rightarrow \boxed{F(\text{leq})} \rightarrow FC \rightarrow \boxed{k} \rightarrow C \\  & R \text{ cost} \sqsubseteq \text{cost} \leq, \text{ as } \text{cost} \text{ is a map; functors}  \end{aligned}  $              |               |
| $  \begin{aligned}  \sqsubseteq & FA \rightarrow \boxed{F(\text{cost})} \rightarrow FC \rightarrow \boxed{k} \rightarrow \boxed{\text{leq}} \rightarrow C \\  & \text{assumption } F(\leq) k \leq k \leq: k \text{ monotonic on } \leq  \end{aligned}  $                                          |               |
| $  \begin{aligned}  = & FA \rightarrow \boxed{h} \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow \boxed{\text{leq}} \rightarrow C \\  & \text{assumption } h \text{ cost} = F(\text{cost}) k  \end{aligned}  $                                                                          |               |

| the condition                                                                                                                                                                                                                       | how it is discharged                                                                 |
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| $F(R)h \sqsubseteq hR$<br>Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $h \text{ cost} = F(\text{cost})k$ , $F(\leq)k \sqsubseteq k \leq$                                                                   | monotonicity when the cost is itself a fold with a step $k$ monotonic on $\leq$      |
| $F(Rn(H^\circ H))h \sqsubseteq hR$<br>Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle)k$ , $F(\leq \times 1)k \sqsubseteq k \leq$ , $H^\circ$ simple; | monotonicity <b>in context</b> : $k$ may also read the input the part was built from |
| $QF(H)h \sqsubseteq F(H)hR$ at $Q \triangleq F(U, V)$<br>Proposition 9.4, $U, V$ preorders, $F(U, R)h \sqsubseteq hR$ , $VH \sqsubseteq HR$                                                                                         | both conditions at once, split along the two arguments of a bifunctor                |

$$\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge R = \text{cost} \leq \text{cost}^\circ \wedge \text{cost} = F(\langle \text{cost}, S \rangle) k \wedge F(\leq \times 1) k \leq k \leq \text{cost} \implies F(R \cap S S^\circ) h \leq h R$$

improving each part by  $R$  within its  $S$ -context, then assembling by  $h$ , is below  $hR$

| circuit                                                                                                                                                                                                                                                                                                                                                                                                             | Hinze–Marsden |
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| $FA \rightarrow \boxed{F(R \cap S S^\circ)} \rightarrow \boxed{h} \rightarrow A$                                                                                                                                                                                                                                                                                                                                    |               |
| $\sqsubseteq FA \rightarrow \boxed{F(R \cap S S^\circ)} \rightarrow \boxed{h} \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A$<br>shunting: $\text{cost}$ a map, so $1 \leq \text{cost} \text{ cost}^\circ$ — triple-chains                                                                                                                               |               |
| $= FA \rightarrow \boxed{F(\langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ)} \rightarrow \boxed{h} \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A$<br>products: $R \cap S S^\circ = \langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ$ at $R = \text{cost} \leq \text{cost}^\circ$ — $\triangleq \times$ |               |
| $= FA \rightarrow \boxed{F(X)} \rightarrow \boxed{F(\langle \text{cost}, S \rangle)} \rightarrow F(C \times B) \rightarrow \boxed{k} \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A$<br>assumption on $\text{cost}$ : $h \text{ cost} = F(\langle \text{cost}, S \rangle) k$                                                                                                                           |               |
| $\sqsubseteq FA \rightarrow \boxed{F(\langle \text{cost} \leq, S \rangle)} \rightarrow F(C \times B) \rightarrow \boxed{k} \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A$<br>$S$ simple, so $\langle \text{cost}, S \rangle^\circ \langle \text{cost}, S \rangle \leq 1$                                                                                                                              |               |
| $= FA \rightarrow \boxed{F(\langle \text{cost}, S \rangle)} \rightarrow \boxed{F(\leq \times 1)} \rightarrow F(C \times B) \rightarrow \boxed{k} \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A$<br>products; functors: $\langle \text{cost} \leq, S \rangle = \langle \text{cost}, S \rangle (\leq \times 1)$ — bdm-prod-laws, relator-laws                                                           |               |
| $\sqsubseteq FA \rightarrow \boxed{F(\langle \text{cost}, S \rangle)} \rightarrow F(C \times B) \rightarrow \boxed{k} \rightarrow \boxed{\leq} \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A$<br>assumption on $k$ : $F(\leq \times 1) k \leq k \leq \text{cost}$                                                                                                                                     |               |

| circuit                                                                                                                                                                                        | Hinze–Marsden |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $= \text{FA} \rightarrow \boxed{h} \rightarrow \boxed{R} \rightarrow A$ <p>assumption on <math>\text{cost}</math> read backwards, then <math>R = \text{cost} \leq \text{cost}^\circ</math></p> |               |

$$G(U, R) h \in h R \wedge V H \in H R \implies G(U, V) G(1, H) h \in G(1, H) h R$$

(15.1.1k)

thinning the parts by  $U$  in the first argument and by  $V$  in the second ( $G(U, V)$ ), then solving by  $H$  and assembling by  $h$ , gives only what solving and assembling and then improving by  $R$  gives ( $G(1, H) h R$ )

| circuit                                                                                                                                                                                                                  | Hinze–Marsden |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $G \ e \ w \rightarrow \boxed{G(U, V)} \rightarrow \boxed{G(1, H)} \rightarrow G \ e \ A \rightarrow \boxed{h} \rightarrow A$                                                                                            |               |
| $= G \ e \ w \rightarrow \boxed{G(U, VH)} \rightarrow G \ e \ A \rightarrow \boxed{h} \rightarrow A$ <p>taking <math>Q \triangleq G(U, V)</math>; bifunctors: <math>G(U, V) G(1, H) = G(U, VH)</math> — relator-laws</p> |               |
| $\sqsubseteq G \ e \ w \rightarrow \boxed{G(U, HR)} \rightarrow G \ e \ A \rightarrow \boxed{h} \rightarrow A$ <p>assumption on <math>V</math>: <math>VH \in HR</math></p>                                               |               |
| $= G \ e \ w \rightarrow \boxed{G(1, H)} \rightarrow \boxed{G(U, R)} \rightarrow G \ e \ A \rightarrow \boxed{h} \rightarrow A$ <p>bifunctors: <math>G(U, HR) = G(1, H) G(U, R)</math> — relator-laws</p>                |               |
| $\sqsubseteq G \ e \ w \rightarrow \boxed{G(1, H)} \rightarrow G \ e \ A \rightarrow \boxed{h} \rightarrow \boxed{R} \rightarrow A$ <p>assumption on <math>h</math>: <math>G(U, R) h \in h R</math></p>                  |               |

## §15.2. The string edit problem

$\text{Op} ::= \text{cpy Char} \mid \text{del Char} \mid \text{ins Char}, \ F(A, B) = 1 + (A \times B), \ \alpha \triangleq [\text{nil}, \text{cons}].$

(15.2a)

$\text{edit} \triangleq ([\text{base}, \text{step}]) : [\text{Op}] \rightarrow [\text{Char}] \times [\text{Char}]$ , base returning  $([], [])$ , step (cpy  $a, (xs, ys)$ ) =  $([a] \# xs, [a] \# ys)$ , step (del  $a, (xs, ys)$ ) =  $([a] \# xs, ys)$ , step (ins  $a, (xs, ys)$ ) =  $(xs, [a] \# ys)$ .

$\text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}])$ ,  $R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U \triangleq \top$ ,  $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$ ,  $Q \triangleq 1 + (U \times V)$ , empty the coreflexive on  $(xs, ys)$  with both lists empty.

unstep implements  $\frac{\text{step}^\circ}{\exists} \text{thin}(U \times V) : \text{unstep } ([a] \# xs, []) = [(\text{del } a, (xs, []))]$ ,  $\text{unstep } ([], [b] \# ys) = [(\text{ins } b, ([], ys))]$ ,  $\text{unstep } ([a] \# xs, [b] \# ys) = (a = b \rightarrow [(\text{cpy } a, (xs, ys))], [(\text{del } a, (xs, [b] \# ys)), (\text{ins } b, ([a] \# xs, ys))])$ .

$(\mu X : \frac{\text{editAlg}^\circ}{\exists} \text{thin}(Q) P(F(X) [\text{nil}, \text{cons}]) \text{est}(R)) \sqsubseteq \frac{\text{edit}^\circ}{\exists} \text{est}(R)$ ,  $\text{mle} = (\text{empty} \rightarrow \text{nil}, \text{unstep list}((1 \times \text{mle}) \text{cons}) \text{minlist}(R))$

(15.2b)

a shortest edit sequence from which both strings can be reconstituted is one pass over the two of them, each step copying, deleting or inserting one character and the best sequence for what is left taken from the entries already computed

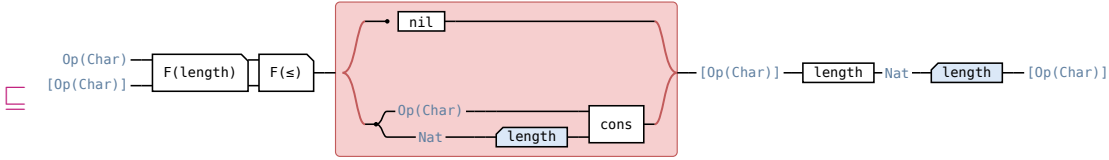
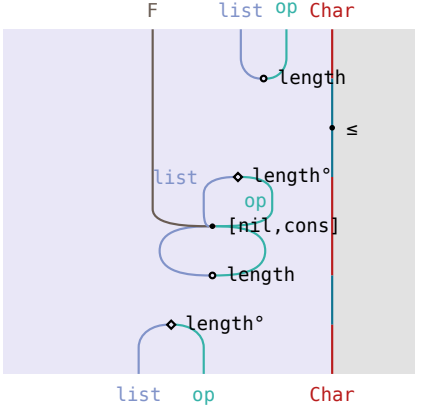
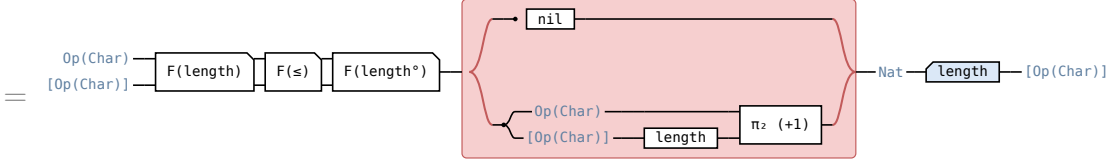
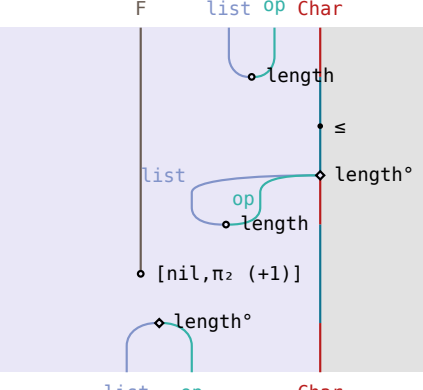
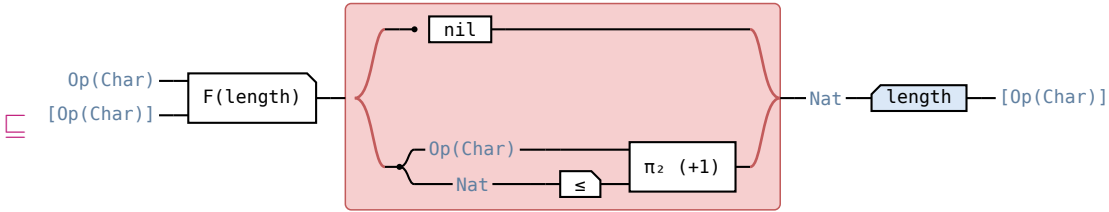
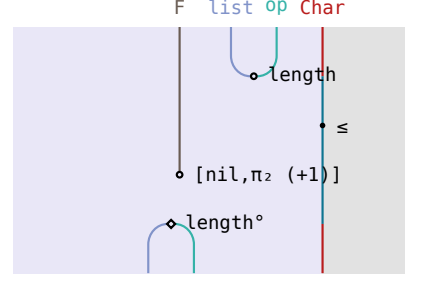
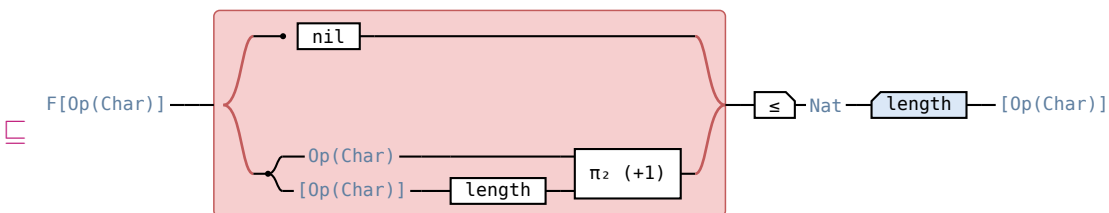
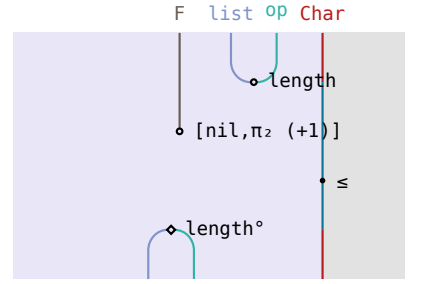
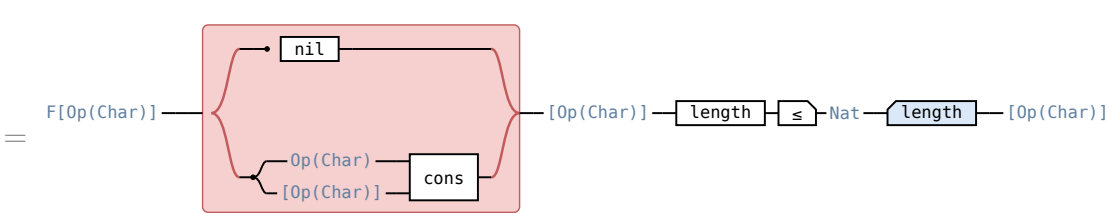
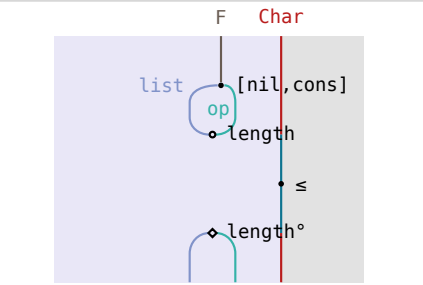
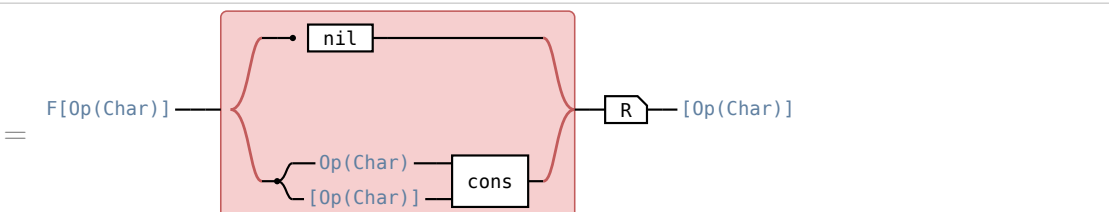
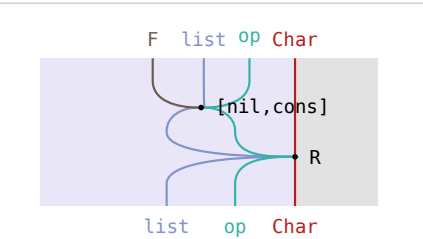
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Hinze–Marsden |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $  \begin{array}{c}  [\text{Char}] \\  [\text{Char}]  \end{array}  \begin{array}{c}  \boxed{\frac{1}{3}} \\  \boxed{\frac{1}{3}}  \end{array}  \begin{array}{c}  E[\text{Char}]^2 \\  E[\text{Char}]^2  \end{array}  \begin{array}{c}  E(\text{edit}^\circ) \\  E(\text{edit}^\circ)  \end{array}  \begin{array}{c}  E[\text{Op}(\text{Char})] \\  E[\text{Op}(\text{Char})]  \end{array}  \begin{array}{c}  \text{est}(\text{R}) \\  \text{est}(\text{R})  \end{array}  \begin{array}{c}  [\text{Op}(\text{Char})] \\  [\text{Op}(\text{Char})]  \end{array}  $ <p>the specification <math>\triangleq \text{edit}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| $  \begin{array}{c}  [\text{Char}] \\  [\text{Char}]  \end{array}  \begin{array}{c}  \boxed{\frac{1}{3}} \\  \boxed{\frac{1}{3}}  \end{array}  \begin{array}{c}  E[\text{Char}]^2 \\  E[\text{Char}]^2  \end{array}  \begin{array}{c}  E(\text{editAlg}^\circ) \\  E(\text{editAlg}^\circ)  \end{array}  \begin{array}{c}  \text{thin}(Q) \\  \text{thin}(Q)  \end{array}  \begin{array}{c}  E[\text{Char}]^2 \\  E[\text{Char}]^2  \end{array}  \begin{array}{c}  P(F(X)[\text{nil}, \text{cons}]) \\  P(F(X)[\text{nil}, \text{cons}])  \end{array}  \begin{array}{c}  E[\text{Op}(\text{Char})] \\  E[\text{Op}(\text{Char})]  \end{array}  \begin{array}{c}  \text{est}(\text{R}) \\  \text{est}(\text{R})  \end{array}  \begin{array}{c}  [\text{Op}(\text{Char})] \\  [\text{Op}(\text{Char})]  \end{array}  $ <p>at <math>Q \triangleq 1 + (U \times V)</math>. Monotonicity <math>F(R) \alpha \sqsubseteq \alpha R</math> is Proposition 9.2 at <math>\text{length} \triangleq ([\text{zero}, \pi_2 \text{succ}])</math> with <math>\text{succ}</math> monotonic on <math>\leq</math>, so <math>\text{cons}</math> is monotonic on <math>R</math>. The thinning condition <math>QF(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R</math> over <math>F(\text{Op}, [\text{Char}] \times [\text{Char}])</math> is Proposition 9.4 at <math>U \triangleq T - F(T, R) \alpha \sqsubseteq \alpha R</math>, left as an exercise, so any two operations may be compared — and <math>V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ</math>, — <math>\text{edit}(1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}</math>, <math>\text{edit}(\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}</math>. <math>\text{suffix} = \text{tail}^*</math> and <math>BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B</math> cut those to one step each: drop the operation that produced the head, or weaken its <math>\text{cpy}</math> to a <math>\text{del}</math>, never lengthening the sequence</p> |               |
| $  \begin{array}{c}  [\text{Char}] \\  [\text{Char}]  \end{array}  \begin{array}{c}  \boxed{\frac{1}{3}} \\  \boxed{\frac{1}{3}}  \end{array}  \begin{array}{c}  E[\text{Char}]^2 \\  E[\text{Char}]^2  \end{array}  \begin{array}{c}  E(\text{step}^\circ) \\  E(\text{step}^\circ)  \end{array}  \begin{array}{c}  \text{thin}(T \times V) \\  \text{thin}(T \times V)  \end{array}  \begin{array}{c}  E[\text{Op}(\text{Char}) \times [\text{Char}] \times [\text{Char}]] \\  E[\text{Op}(\text{Char}) \times [\text{Char}] \times [\text{Char}]]  \end{array}  \begin{array}{c}  P((1 \times X) \text{cons}) \\  P((1 \times X) \text{cons})  \end{array}  \begin{array}{c}  E[\text{Op}(\text{Char})] \\  E[\text{Op}(\text{Char})]  \end{array}  \begin{array}{c}  \text{est}(\text{R}) \\  \text{est}(\text{R})  \end{array}  \begin{array}{c}  [\text{Op}(\text{Char})] \\  [\text{Op}(\text{Char})]  \end{array}  $ <p><math>\equiv</math> Proposition 9.1: <b>base</b> and <b>step</b> have disjoint ranges, so the fixed point splits into one branch per summand — <b>empty</b>, the coreflexive on <math>(xs, ys)</math> with both lists empty, is where <b>base</b> returns</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| $  \begin{array}{c}  [\text{Char}] \\  [\text{Char}]  \end{array}  \begin{array}{c}  \boxed{\frac{1}{3}} \\  \boxed{\frac{1}{3}}  \end{array}  \begin{array}{c}  E[\text{Char}]^2 \\  E[\text{Char}]^2  \end{array}  \begin{array}{c}  E(\text{step}^\circ) \\  E(\text{step}^\circ)  \end{array}  \begin{array}{c}  \text{thin}(T \times V) \\  \text{thin}(T \times V)  \end{array}  \begin{array}{c}  E[\text{Op}(\text{Char}) \times [\text{Char}] \times [\text{Char}]] \\  E[\text{Op}(\text{Char}) \times [\text{Char}] \times [\text{Char}]]  \end{array}  \begin{array}{c}  P((1 \times X) \text{cons}) \\  P((1 \times X) \text{cons})  \end{array}  \begin{array}{c}  E[\text{Op}(\text{Char})] \\  E[\text{Op}(\text{Char})]  \end{array}  \begin{array}{c}  \text{est}(\text{R}) \\  \text{est}(\text{R})  \end{array}  \begin{array}{c}  [\text{Op}(\text{Char})] \\  [\text{Op}(\text{Char})]  \end{array}  $ <p><math>\equiv</math> <b>unstep</b> implements <math>\frac{\text{step}^\circ}{3} \text{thin}(U \times V)</math> — at most two decompositions survive, a copy beating a delete or an insert wherever it is available — and <b>minlist</b>(R) implements <b>est</b>(R). The same subproblem is solved many times over, so the running time is exponential in the two lengths</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |

$$F(R)[\text{nil}, \text{cons}] \sqsubseteq [\text{nil}, \text{cons}]R$$

comparing the tails by  $R$  and then putting an operation in front ( $F(R)\alpha$ ) relates only sequences that  $R$  also relates once the operation is in front ( $\alpha R$ )

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Hinze–Marsden |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $  \begin{array}{c}  [\text{Op}(\text{Char})] \\  [\text{Op}(\text{Char})]  \end{array}  \begin{array}{c}  \boxed{\text{nil}} \\  \boxed{\text{nil}}  \end{array}  \begin{array}{c}  \text{Op}(\text{Char}) \\  \text{Op}(\text{Char})  \end{array}  \begin{array}{c}  \text{R} \\  \text{R}  \end{array}  \begin{array}{c}  \text{cons} \\  \text{cons}  \end{array}  \begin{array}{c}  [\text{Op}(\text{Char})] \\  [\text{Op}(\text{Char})]  \end{array}  $                                                                                                                                                                                                                      |               |
| $  \begin{array}{c}  \text{Op}(\text{Char}) \\  [\text{Op}(\text{Char})]  \end{array}  \begin{array}{c}  \boxed{\text{length}} \\  \boxed{\text{length}}  \end{array}  \begin{array}{c}  \text{F}(\leq) \\  \text{F}(\leq)  \end{array}  \begin{array}{c}  \text{Op}(\text{Char}) \\  \text{Op}(\text{Char})  \end{array}  \begin{array}{c}  \text{length} \\  \text{length}  \end{array}  \begin{array}{c}  \text{cons} \\  \text{cons}  \end{array}  \begin{array}{c}  [\text{Op}(\text{Char})] \\  [\text{Op}(\text{Char})]  \end{array}  $ <p><math>R \triangleq \text{length} \leq \text{length}^\circ \triangleq \text{edit}</math>; <math>F</math> preserves composition</p> |               |

(15.2c)

| circuit                                                                                                                                                                                                                                                               | Hinze–Marsden                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <p><math>\text{length}</math> is a map, so entire: <math>1 \sqsubseteq \text{length} \text{ length}^\circ</math></p>                                                               |    |
|  <p><math>\alpha \text{ length} = F(\text{length})[\text{zero}, \pi_2 \text{ succ}]: \text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}]) \triangleq \text{edit}</math></p> |    |
|  <p><math>\text{length}</math> is a map, so simple, and <math>F</math> is monotonic: <math>F(\text{length}^\circ)F(\text{length}) \sqsubseteq F(1) = 1</math></p>                 |  |
|  <p><math>\text{SUCC}</math> is monotonic on <math>\leq</math>: <math>F(\leq)[\text{zero}, \pi_2 \text{ succ}] \sqsubseteq [\text{zero}, \pi_2 \text{ succ}] \leq</math></p>      |  |
|  <p><math>F(\text{length})[\text{zero}, \pi_2 \text{ succ}] = \alpha \text{ length}</math> again</p>                                                                              |  |
|  <p><math>R \triangleq \text{length} \leq \text{length}^\circ \triangleq \text{edit}</math></p>                                                                                   |  |

$$QF(\text{edit}^\circ)[\text{nil}, \text{cons}] \sqsubseteq F(\text{edit}^\circ)[\text{nil}, \text{cons}]R$$

every edit sequence built from a decomposition  $Q$  puts above a given one — any operation, each string lengthened at the front — is at least as long as one built from the given one

(15.2d)

| circuit                                                                                                                                                                                                                                       | Hinze–Marsden |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|                                                                                                                                                                                                                                               |               |
| $Q \triangleq 1 + (U \times V) \text{ is } F(U, V) \text{ at } U \triangleq \tau \text{ — } \triangleq \text{edit}; \text{ and the bifunctor } F \text{ preserves composition: } F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)$ |               |
| $Q \triangleq 1 + (U \times V) \text{ is } F(U, V) \text{ at } U \triangleq \tau \text{ — } \triangleq \text{edit}; \text{ and the bifunctor } F \text{ preserves composition: } F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)$ |               |
| $Q \triangleq 1 + (U \times V) \text{ is } F(U, V) \text{ at } U \triangleq \tau \text{ — } \triangleq \text{edit}; \text{ and the bifunctor } F \text{ preserves composition: } F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)$ |               |
| $Q \triangleq 1 + (U \times V) \text{ is } F(U, V) \text{ at } U \triangleq \tau \text{ — } \triangleq \text{edit}; \text{ and the bifunctor } F \text{ preserves composition: } F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)$ |               |

(15.2e)

| <b>edit <math>V^\circ \sqsubseteq R^\circ \text{ edit}</math></b><br>cutting a front off either string an edit sequence produces ( <b>edit <math>V^\circ</math></b> ) leaves a pair that a sequence no longer produces ( <b><math>R^\circ</math> edit</b> )                                      |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                          | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                  |               |
| $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ, \text{ and } \times \text{ preserves composition}$                                                                                                                                                                                 |               |
| $\text{edit} (\text{suffix} \times 1) \sqsubseteq R^\circ \text{ edit: drop the operation that produced the head, or weaken its cpy to an ins. Proved by induction at suffix directly, not through suffix=tail* and } BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B \text{ as the book does}$ |               |

| circuit                                                                                                                                                    | Hinze–Marsden |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <p><math>\text{edit } (1 \times \text{suffix}) \subseteq R^\circ \text{ edit}</math>, the mirror image: <code>cpy</code> weakens to a <code>del</code></p> |               |
| <p><math>R^\circ</math> is transitive, <math>\leq</math> being so</p>                                                                                      |               |

| <b>step</b> $\text{base}^\circ = \emptyset$<br>no pair of strings is both a result of <b>step</b> and the result $([], [])$ of <b>base</b>                             |               | (15.2f) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------|
| circuit                                                                                                                                                                | Hinze–Marsden |         |
|                                                                                                                                                                        |               |         |
| <p><math>\text{base}</math> returns only <math>([], [])</math>, so <math>\text{base}^\circ = \text{empty } \text{base}^\circ</math></p>                                |               |         |
| <p><math>\text{step } \text{empty} = \emptyset</math>: <code>cpy</code> and <code>del</code> put a character on the left string, <code>ins</code> one on the right</p> |               |         |
| <p><math>\emptyset</math> composed with anything is <math>\emptyset</math></p>                                                                                         |               |         |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\text{mle}(xs, ys) = \text{head}(\text{column}(xs, ys))$ , $\text{column}(xs, ys) = [\text{mle}(u, ys) \mid u \leftarrow \text{tails}(xs)]$<br>$\text{column}(xs) = ([\text{fstcol}(xs), \text{nextcol}(xs)] \Downarrow, \text{fstcol} = \text{list}(\text{del}) \text{ tails}$<br>the tabulation: $\text{mle}(xs, ys)$ needs $\text{mle}(u, v)$ for every tail $u$ of $xs$ and $v$ of $ys$ , so the columns are built right to left<br>$\text{column}(xs, \text{cons}(b, ys)) = \text{nextcol}(xs, (b, \text{column}(xs, ys)))$<br>$\text{nextcol}(xs)(b, us) = ([\text{base}(b, \text{last}(us)), \text{step}(b)] \Downarrow)(xus)$ , $xus = \text{zip}(xs, \text{zip}(\text{init}(us), \text{tail}(us)))$<br>each column is a fold built bottom to top, over $xs$ zipped with the adjacent pairs of the column to its right<br>$\text{base}(b, u) = [[\text{ins}(b)] \# u]$<br>$\text{step}(b)((a, (u, v)), ws) = (a \rightarrow b \rightarrow [[\text{cpy}(a)] \# v] \# ws, [\text{bmin}(R)([\text{del}(a)] \# w, [\text{ins}(b)] \# u)] \# ws)$<br>$w = \text{head}(ws)$ ; these $\text{base}$ , $\text{step}$ are not $\text{edit}$ 's. An entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths | (15.2g) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|

### §15.3. Optimal bracketing

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| tree $A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$ , $\text{FX} = A + X^2$ , so $F(R) = 1 + R^2$ ; $h \triangleq [\text{tip}, \text{bin}]$ , $\text{flatten} \triangleq ([\text{wrap}, \text{cat}] \Downarrow) : \text{tree } A \rightarrow \text{list}^+ A$ , $H = \text{flatten}^\circ$ .<br>$\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}] \Downarrow)$ , $\text{opt} \triangleq (\text{zero}, \text{st})$ , $\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))$ .<br>$\text{sb}$ associative, so $\text{size} = \text{flatten } \text{sz}$ for a map $\text{sz} : R \triangleq \text{cost} \leq \text{cost}^\circ$ , $g \triangleq [\text{zero}, (1 \times \text{sz})^2 \text{ opb } \pi_1]$ , single the coreflexive on singleton lists. | (15.3a) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|

$\text{split} \triangleq (\text{inits}^+, \text{tails}^+) \text{ zip}$ , an implementation of  $\frac{\text{cat}^0}{\text{}};$   $\text{array} \triangleq \text{inits list}(\text{row})$ ,  $\text{row} \triangleq \text{tails list}(\text{mct})$ ,  $\text{col} \triangleq \text{inits list}(\text{mct})$ .

$\text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist}(\text{R})$ ,  $\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}$ ,  $\text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})$ .

(15.3b)

| $\text{Assoc}(\text{sb}) \Rightarrow F(\langle \text{cost}, \text{flatten} \rangle)g = [\text{tip}, \text{bin}] \text{cost}$<br>pairing each subtree with its cost and its flattening and then applying $g$ gives what building the tree by $[\text{tip}, \text{bin}]$ and taking its <b>cost</b> gives |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                 | Hinze–Marsden |
| $F(\text{tree}(A)) \rightarrow F(\langle \text{cost}, \text{flatten} \rangle) \rightarrow F(\text{Int} \times \text{list}^+(A)) \rightarrow g \rightarrow \text{Int}$                                                                                                                                   |               |
| $=$ <p>definition of <math>g</math>; coproducts and products <math>\triangleq \text{flatten}</math></p>                                                                                                                                                                                                 |               |
| $=$ <p><math>\text{flatten sz} = \text{size}</math>, since <math>\text{sb}</math> is associative <math>\triangleq \text{flatten}</math></p>                                                                                                                                                             |               |
| $=$ <p><math>\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])</math>, at a node <math>\text{bin}(\text{cost}, \text{size}) = (\text{cost}, \text{size})^2 \text{ opb}</math></p>                                                                                          |               |
| $=$ <p><math>\text{tip cost} = \text{zero}</math>; products, <math>\langle \text{cost}, \text{size} \rangle \pi_1 = \text{cost}</math></p>                                                                                                                                                              |               |
| $=$ <p>coproducts, <math>[\text{tip cost}, \text{bin cost}] = [\text{tip}, \text{bin}] \text{cost}</math></p>                                                                                                                                                                                           |               |

(15.3c)

| $F(\leq \times 1)g \leq g$<br>raising the cost of either subtree ( $\leq \times 1$ in both slots) and then applying $g$ returns a value at least what $g$ returns on the costs before raising ( $g \leq$ ) |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                    | Hinze–Marsden |
| $F(\text{Int} \times \text{list}^+(A)) \rightarrow F(\leq \times 1) \rightarrow g \rightarrow \text{Int}$                                                                                                  |               |

| circuit                                                                                                                                                                                                            | Hinze–Marsden |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ <p>definition of <math>g \triangleq \text{flatten}</math></p>                                            |               |
| $A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ <p>definition of <math>\text{opb}</math>, and <math>+</math> monotonic</p>                               |               |
| $A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ <p><math>\leq</math> reflexive, so <math>\text{zero} \sqsubseteq \text{zero} \leq</math>; coproducts</p> |               |
| $F(\text{Int} \times \text{list}^+(A)) \rightarrow \text{Int}$ <p>definition of <math>g \triangleq \text{flatten}</math></p>                                                                                       |               |

(15.3d)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $\text{Assoc}(\text{sb}) \Rightarrow (\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{\exists} P([\text{tip}, (X \times X) \text{bin}]) \text{est}(R)) \sqsubseteq \frac{\text{flatten}^\circ}{\exists} \text{est}(R), \text{ mct} = (\text{single-head tip}, (\text{init col}, \text{tail row}) \text{mix})$ <p>a least-cost bracketing of <math>a_1 \oplus \dots \oplus a_n</math> is read off an array holding one best tree per non-empty segment, each entry built from the column to its left and the row below it</p> |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Hinze–Marsden |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\text{list}^+(A) \rightarrow \frac{1}{\exists} \text{E}(\text{list}^+(A)) \rightarrow \text{E}(\text{flatten}^\circ) \rightarrow \text{E}(\text{tree}(A)) \rightarrow \text{est}(R) \rightarrow \text{tree}(A)$ <p>the specification <math>\triangleq \text{flatten}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| $\text{list}^+(A) \rightarrow \frac{1}{\exists} \text{E}(\text{list}^+(A)) \rightarrow \text{E}([\text{wrap}, \text{cat}]^\circ) \rightarrow \text{E}(\text{list}^+(A)) \rightarrow P([\text{tip}, (X \times X) \text{bin}]) \rightarrow \text{E}(\text{tree}(A)) \rightarrow \text{est}(R) \rightarrow \text{tree}(A)$ <p>split the list in every way, bracket both halves, join. The condition is monotonicity <b>in context</b>, <math>F(Rn(\text{flatten} \text{ flatten}^\circ)) \text{h} \sqsubseteq \text{h} R</math> — only trees with the same flattening are compared — which is Proposition 9.3 at <math>H^\circ = \text{flatten}</math>, a map, with (9.5) <math>[\text{tip}, \text{bin}]</math><br/> <math>\text{cost} = (1 + (\text{cost}, \text{flatten})^2) g</math> (the cost of a node reads only the cost and the flattening of its two subtrees) and (9.6) <math>(1 + (\leq \times 1)^2) g \sqsubseteq g \leq</math> (<math>g</math> monotonic on <math>\leq</math> in its two cost arguments)</p> |               |
| $\text{list}^+(A) \rightarrow \frac{1}{\exists} \text{E}(\text{list}^+(A)) \rightarrow \text{E}(\text{cat}^\circ) \rightarrow \text{E}(\text{list}^+(A)^2) \rightarrow P((X \times X) \text{bin}) \rightarrow \text{E}(\text{tree}(A)) \rightarrow \text{est}(R) \rightarrow \text{tree}(A)$ <p>Proposition 9.1: <b>wrap</b> and <b>cat</b> have disjoint ranges, and <b>single</b> is the coreflexive on singleton lists, where <b>wrap</b> returns</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| $\text{list}^+(A) \rightarrow \text{splits} \rightarrow [\text{list}^+(A)^2] \rightarrow \text{list}((\text{mct} \times \text{mct}) \text{bin}) \rightarrow [\text{tree}(A)] \rightarrow \text{minlist}(R) \rightarrow \text{tree}(A)$ <p><math>\text{splits} \triangleq (\text{inits}^+, \text{tails}^+)</math> <b>zip</b> implements <math>\frac{\text{cat}^\circ}{\exists}</math> and <math>\text{minlist}(R)</math> implements <math>\text{est}(R)</math>. Exponential, since the segments of one list overlap</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\text{mct} = (\text{single} \rightarrow \text{head tip}, (\text{init col}, \text{tail row}) \text{ mix}) \quad (9.7)$ <p>= the tabulation: <math>\text{mct}</math> <math>\times S</math> is needed for every non-empty segment <math>XS</math>, so the values are held as an array of rows, <math>\text{array} \triangleq \text{inits list}(\text{row}), \text{row} \triangleq \text{tails list}(\text{mct}), \text{col} \triangleq \text{inits list}(\text{mct}), \text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist}(R)</math></p>                                                                     |               |
| $\text{col} = (\text{single} \rightarrow \text{head tip wrap}, (\text{init col}, \text{tail row}) \text{ next}) \quad (9.8)$ $\text{cons col} = (1 \times \text{array}) \text{ process} \quad (9.9)$ <p>= <math>\text{row} = (\text{single} \rightarrow \text{head tip wrap}, (\text{mct}, \text{tail row}) \text{ cons}) \quad (9.10)</math><br/> a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop,<br/> <math>\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}, \text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})</math></p> |               |
| $\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip wrap wrap}$ <p>= <math>\text{addcol} \triangleq (\pi_1 \text{ tip wrap}, \text{step}) \text{ cons}, \text{step} \triangleq (\text{process tail}, \pi_2) \text{ zip list}(\text{cons})</math><br/> the program: one fold building the array column by column, cubic in the length of the input</p>                                                                                                                                                                                                                                  |               |

(15.3e)

| <b>nonsingle mct = nonsingle (init col, tail row) mix</b><br>on a list of two or more elements, <b>mct</b> gives what <b>mix</b> gives on the column of its <b>init</b> and the row of its <b>tail</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
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| $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{mct} \rightarrow \text{tree}(A)$ <p>= <math>\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{splits} \rightarrow [\text{list}^+(A)^2] \rightarrow \text{list}(\text{graft}) \rightarrow [\text{tree}(A)] \rightarrow \text{minlist}(R) \rightarrow \text{tree}(A)</math> recursive case of <math>\text{mct} \triangleq \text{flatten}</math></p> <p>= <math>\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{inits}^+ \\ \text{tails}^+ \end{cases} \rightarrow \begin{cases} [\text{list}^+(A)] \\ [\text{list}^+(A)] \end{cases} \rightarrow \text{zip} \rightarrow [\text{list}^+(A)^2] \rightarrow \text{list}(\text{graft}) \rightarrow [\text{tree}(A)] \rightarrow \text{minlist}(R) \rightarrow \text{tree}(A)</math><br/> definition of <math>\text{splits} \triangleq \text{flatten}</math></p> <p>= <math>\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{trees} \rightarrow \text{zip} \rightarrow [\text{tree}(A)^2] \rightarrow \text{list}(\text{bin}) \rightarrow [\text{tree}(A)] \rightarrow \text{minlist}(R) \rightarrow \text{tree}(A)</math><br/> definitions of <math>\text{graft} \triangleq (\text{mct} \times \text{mct})</math> <math>\text{bin}</math> and <math>\text{trees} \triangleq (\text{inits}^+ \text{ list}(\text{mct}), \text{tails}^+ \text{ list}(\text{mct}))</math>, our names; <math>\text{zip list}(f \times g) = (\text{list}(f) \times \text{list}(g)) \text{ zip}</math></p> <p>= <math>\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{trees} \rightarrow \text{mix} \rightarrow \text{tree}(A)</math> introducing <math>\text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist}(R) \triangleq \text{flatten}</math></p> <p>= <math>\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{init} &amp; \text{col} \\ \text{tail} &amp; \text{row} \end{cases} \rightarrow \begin{cases} [\text{tree}(A)] \\ [\text{tree}(A)] \end{cases} \rightarrow \text{mix} \rightarrow \text{tree}(A)</math><br/> definition of <math>\text{trees}</math>; <math>\text{inits}^+ = \text{init inits}</math> and <math>\text{tails}^+ = \text{tail tails}</math> on non-singletons; definition of <math>\text{row}</math> and <math>\text{col} \triangleq \text{flatten}</math></p> |  |

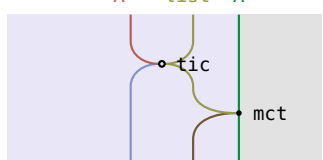
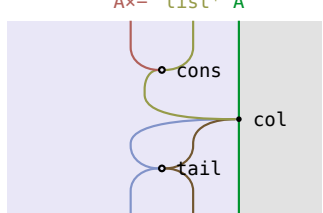
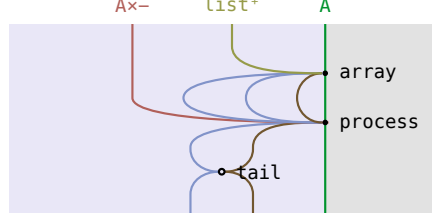
(15.3f)

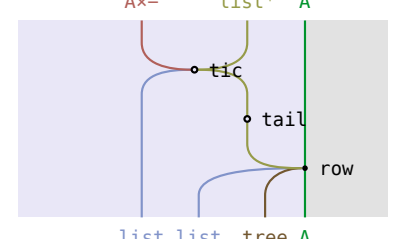
| <b>nonsingle col = nonsingle (init col, tail row) next</b><br>on a list of two or more elements, the column is what <b>next</b> makes of the column of its <b>init</b> and the row of its <b>tail</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{col} \rightarrow [\text{tree}(A)]$ <p>= <math>\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{inits} \rightarrow [\text{list}^+(A)] \rightarrow \text{list}(\text{mct}) \rightarrow [\text{tree}(A)]</math> definition of <math>\text{col} \triangleq \text{flatten}</math></p> <p>= <math>\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{init} &amp; \text{inits} \\ &amp; \text{list}^+(A) \end{cases} \rightarrow \text{snoc} \rightarrow [\text{list}^+(A)] \rightarrow \text{list}(\text{mct}) \rightarrow [\text{tree}(A)]</math><br/> <math>\text{inits} = (\text{init inits}, 1)</math> <math>\text{snoc}</math> on non-singletons</p> <p>= <math>\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{init} &amp; \text{col} \\ \text{mct} &amp; \text{tree}(A) \end{cases} \rightarrow \text{snoc} \rightarrow [\text{tree}(A)]</math> <math>\text{snoc list}(f) = (\text{list}(f) \times f) \text{ snoc}</math>; definition of <math>\text{col}</math></p> <p>= <math>\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{init} &amp; \text{col} \\ \text{tail} &amp; \text{row} \end{cases} \rightarrow \begin{cases} [\text{tree}(A)] \\ [\text{tree}(A)] \end{cases} \rightarrow \text{next} \rightarrow [\text{tree}(A)]</math><br/> (9.7) on non-singletons — (15.3e) — and introducing <math>\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc} \triangleq \text{flatten}</math></p> |  |

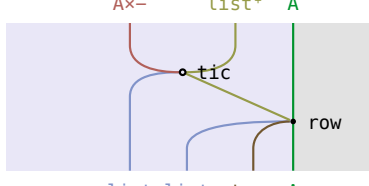
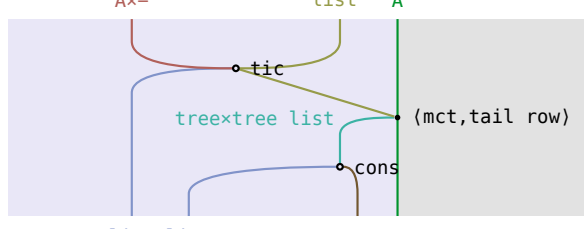
(15.3g)

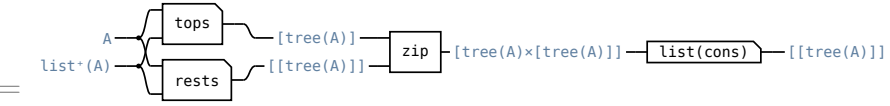
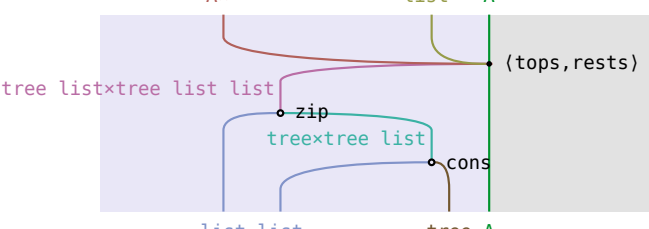
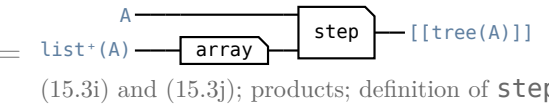
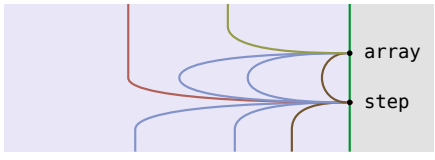
| <b>nonsingle row = nonsingle (mct, tail row) cons</b><br>on a list of two or more elements, the row is <b>mct</b> of the whole list consed onto the row of its <b>tail</b> |  |
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| circuit                                                                                                                                                                                                                                                                                                                                                  | Hinze–Marsden                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{inits}} \text{---} \boxed{\text{tail}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list(mct)}} \text{---} [\text{tree}(A)]$ <p>definition of <math>\text{tops} \triangleq \text{tic}</math> <math>\text{list(mct)}</math>, our name</p> |  |
| $= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{col}} \text{---} \boxed{\text{tail}} \text{---} [\text{tree}(A)]$ <p><math>\text{tail list}(f) = \text{list}(f) \text{ tail}</math>; definition of <math>\text{col}</math></p>                                                                    |  |
| $= \text{list}^+(A) \text{---} \boxed{\text{array}} \text{---} \boxed{\text{process}} \text{---} \boxed{\text{tail}} \text{---} [\text{tree}(A)]$ <p>(9.9) — (15.3h)</p>                                                                                                                                                                                 |  |

| <b>rests = <math>\pi_2</math> array</b><br>the rest of each new row is the array of $\mathbf{x}$                                                                                                                                                                                                                                                                      |                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                                                                                                                                                               | Hinze–Marsden                                                                         |
| $\text{list}^+(A) \text{---} \boxed{\text{rests}} \text{---} [[\text{tree}(A)]]$                                                                                                                                                                                                                                                                                      |  |
| $= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{inits}} \text{---} \boxed{\text{tail}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list(tail row)}} \text{---} [[\text{tree}(A)]]$ <p>definition of <math>\text{rests} \triangleq \text{tic}</math> <math>\text{list(tail row)}</math>, our name</p> |  |
| $= \text{list}^+(A) \text{---} \text{list}^+(A) \text{---} \boxed{\text{array}} \text{---} [[\text{tree}(A)]]$ <p><math>\text{tic list(tail)} = \pi_2 \text{ inits}</math>; definition of <math>\text{array}</math></p>                                                                                                                                               |  |

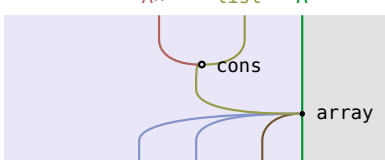
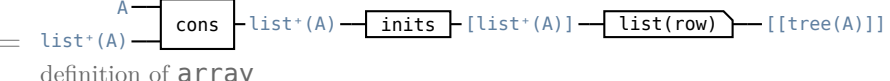
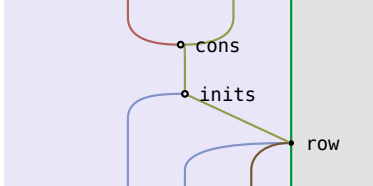
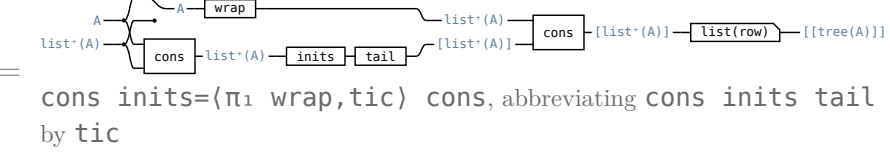
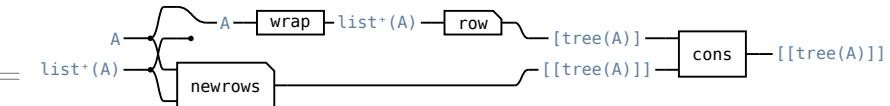
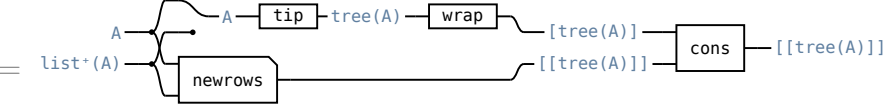
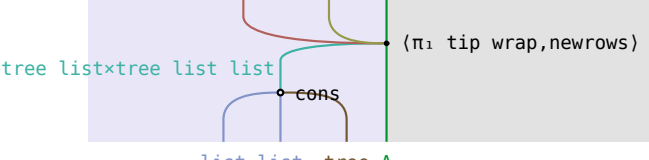
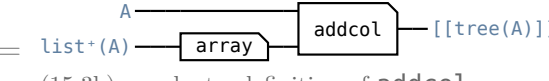
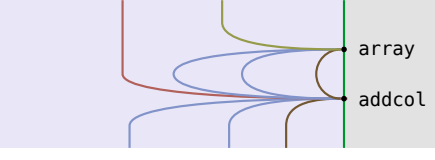
| <b>newrows = <math>(1 \times \text{array}) \text{step}</math></b><br>the rows of the array of $\mathbf{a}:\mathbf{x}$ below its first are what <b>step</b> makes of $\mathbf{a}$ and the array of $\mathbf{x}$                                                                                                                                                |                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                                                                                                                                                       | Hinze–Marsden                                                                         |
| $\text{list}^+(A) \text{---} \boxed{\text{newrows}} \text{---} [[\text{tree}(A)]]$                                                                                                                                                                                                                                                                            |  |
| $= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{inits}} \text{---} \boxed{\text{tail}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list(row)}} \text{---} [[\text{tree}(A)]]$ <p>definition of <math>\text{newrows} \triangleq \text{tic}</math> <math>\text{list(row)}</math>, our name</p> |  |
| $= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{inits}} \text{---} \boxed{\text{tail}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list(mct, tail row)cons}} \text{---} [[\text{tree}(A)]]$ <p>(9.10) — (15.3g) — on the non-singleton <math>\text{tic}</math> lists</p>                     |  |

| circuit                                                                                                                                                                                                                       | Hinze–Marsden                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  <p><math>\text{list}(f,g) = (\text{list}(f), \text{list}(g)) \text{ zip}; \text{products}; \text{definitions of tops and rests}</math></p> |  |
|  <p><math>\text{list}^+(A) \text{ array step } [[\text{tree}(A)]]</math><br/>(15.3i) and (15.3j); products; definition of <b>step</b></p>    |  |

### cons array=(1xarray)addcol

the array of **a** consed onto **x** is what **addcol** makes of **a** and the array of **x**

(15.3l)

| circuit                                                                                                                                                                                                                | Hinze–Marsden                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <p><math>\text{list}^+(A) \text{ cons array } [[\text{tree}(A)]]</math><br/>definition of <b>array</b></p>                           |    |
|  <p><math>\text{list}^+(A) \text{ cons inits list(row) } [[\text{tree}(A)]]</math><br/>definition of <b>array</b></p>              |  |
|  <p><math>\text{list}^+(A) \text{ cons inits tail list(row) } [[\text{tree}(A)]]</math><br/>definition of <b>array</b></p>         |  |
|  <p><math>\text{list}^+(A) \text{ wrap list}^+(A) \text{ row cons } [[\text{tree}(A)]]</math><br/>definition of <b>newrows</b></p> |  |
|  <p><math>\text{list}^+(A) \text{ tip tree}(A) \text{ wrap cons } [[\text{tree}(A)]]</math><br/>definition of <b>newrows</b></p>   |  |
|  <p><math>\text{list}^+(A) \text{ array addcol } [[\text{tree}(A)]]</math><br/>(15.3k); products; definition of <b>addcol</b></p>   |  |

## §15.4. Data compression

$[A] ::= \text{nil} | \text{snoc } ([A], A), \text{ list}^+ A ::= \text{wrap } A | \text{snoc } (\text{list}^+ A, A), \text{ String} = [\text{Char}], \text{ Code} ::= \text{sym } \text{Char} | \text{ptr } (\text{String}, \text{String}^+), \text{ F}(\text{Code}, \text{String}) = 1 + (\text{String} \times \text{Code}), \alpha \triangleq [\text{nil}, \text{snoc}].$

(15.4a)

$\text{decode} \triangleq ([\text{nil}, \text{extend}]) : [\text{Code}] \rightarrow \text{String}, \text{ extend } (xs, \text{sym } a) = xs \# [a], \text{ extend } (xs, \text{ptr } (ys, zs)) = xs \# zs \text{ when } ys \# zs \text{ is a proper prefix of } xs \# zs; \text{ H} = \text{decode}^\circ.$

$\text{size} \triangleq ([\text{zero}, \text{distr } [1 \times c, 1 \times p] \text{ plus}])$  with  $c, p$  the constant costs of a symbol and a pointer;  $R \triangleq \text{size} \leq \text{size}^\circ$ ,  $Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$ , the two  $\tau$  on symbols and on pointers.

$\text{lrt } ws = \text{est}(\text{prefix}^\circ \times (\tau + \tau)) \{ (xs, (ys, zs)) \mid xs \# zs = ws, ys \# zs \text{ a proper prefix of } ws \}$ , the longest repeated tail;  $\text{reduce } (ws \# [a]) = (zs \# [] \rightarrow [(ws, \text{sym } a), (xs, \text{ptr } (ys, zs))], [(ws, \text{sym } a)])$  with  $(xs, (ys, zs)) = \text{lrt } (ws \# [a])$ .

$$0 \leq c \wedge 0 \leq p \Rightarrow (\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{3} \text{thin}(Q) P([\text{nil}, (X \times 1) \text{snoc}]) \text{est}(R)) \sqsubseteq \frac{\text{decode}^\circ}{3} \text{est}(R), \text{ encode} = (\text{null} \rightarrow \text{nil}, \text{reduce list}((\text{encodex1}) \text{snoc}) \text{minlist}(R))$$

(15.4b)

a smallest code sequence decoding to the given string is built from the right, each step emitting the last character as a symbol or ending with a pointer back into what has already been decoded

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Hinze–Marsden |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <p>the specification — <math>\triangleq</math> decode</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <p>at <math>Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))</math>, the two <math>\tau</math> on symbols and on pointers. Monotonicity <math>F(R) \alpha \sqsubseteq \alpha R</math> is routine, the two costs being constants: <math>(\tau + \tau)[c, p] = [c, p]</math>.</p> <p>□ Proposition 9.4 splits the thinning condition in two — <math>F(\tau + \tau, R) \alpha \sqsubseteq \alpha R</math>, and <math>\text{decode prefix} \sqsubseteq R^\circ \text{ decode}</math>, for which <math>\text{decode init} \sqsubseteq R^\circ \text{ decode}</math> is enough: dropping the last character of the output shortens or removes the last code element and never raises the cost. So between a symbol and a pointer nothing can be decided in advance, and between two pointers the longer match wins</p> |               |
| <p>= Proposition 9.1: <math>\text{nil}</math> and <math>\text{extend}</math> have disjoint ranges. The decompositions of one string are <math>\frac{\text{extend}^\circ}{3} (ws \# [a]) = \{(ws, \text{sym } a)\} \cup \{(xs, \text{ptr } (ys, zs)) \mid xs \# zs = ws \# [a], ys \# zs \text{ a proper prefix of } ws\}</math> — take the last character as a symbol, or end with a pointer</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <p>□ <math>\text{reduce}</math> implements <math>\frac{\text{extend}^\circ}{3} \text{thin}(\text{prefix}^\circ \times (\tau + \tau))</math>: thinning leaves at most two, the symbol and the pointer of the longest repeated tail <math>\text{lrt}</math>. Again exponential; the book gives no tabulation for it</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |

$$0 \leq c \wedge 0 \leq p \Rightarrow QF(\text{decode}^\circ)[\text{nil}, \text{snoc}] \sqsubseteq F(\text{decode}^\circ)[\text{nil}, \text{snoc}]R$$

(15.4c)

every code sequence built from a decomposition  $Q$  puts above a given one — a code element of the same kind, a longer front string — costs at least as much as one built from the given one

| circuit | Hinze–Marsden |
|---------|---------------|
|         |               |

| circuit                                                                                                                                                                                                                                                                       | Hinze–Marsden |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <p><math>Q \triangleq 1 + (\text{prefix}^\circ \times (T+T))</math> is <math>F(T+T, \text{prefix}^\circ) \triangleq \text{decode}</math></p>                                                                                                                                  |               |
| <p>the bifunctor <math>F</math> preserves composition: <math>F(U, \text{prefix}^\circ) F(1, \text{decode}^\circ) = F(U, \text{prefix}^\circ \text{decode}^\circ)</math></p>                                                                                                   |               |
| <p>Proposition 9.4's second condition <math>\text{prefix}^\circ \text{decode}^\circ \sqsubseteq \text{decode}^\circ R</math>: dropping the last <b>sym</b>, or shortens or drops the last pointer. Proved at <b>prefix</b> directly by induction, not through <b>init</b></p> |               |
| <p><math>F</math> preserves composition: <math>F(U, \text{decode}^\circ R) = F(1, \text{decode}^\circ) F(U, R)</math></p>                                                                                                                                                     |               |
| <p>Proposition 9.4's first condition <math>F(T+T, R) \alpha \sqsubseteq \alpha R</math>, left as an exercise in the book: <b>snoc</b> adds the cost of the last element to both sides, and <math>[c, p] (T+T) = [c, p]</math></p>                                             |               |

| <div> <b>extend nil°=0</b><br/>                     no string is both a result of <b>extend</b> and the result <b>[]</b> of <b>nil</b> </div> |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                       | Hinze–Marsden |
|                                                                                                                                               |               |
| <p><math>\text{nil}</math> returns only <b>[]</b>, so <math>\text{nil}^\circ = \text{null } \text{nil}^\circ</math></p>                       |               |
| <p><b>extend null=0</b>: a symbol ends the string with a character, a pointer with its non-empty <b>zs</b></p>                                |               |

(15.4d)

| circuit                                                                                                                                                                                                                                                                                                                                                      | Hinze–Marsden                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="138 186 314 247"><div data-bbox="182 186 249 213">[Char]</div><div data-bbox="138 226 249 247">= Code</div></div> <div data-bbox="262 186 314 247"><div data-bbox="262 186 314 247" style="border: 1px solid black; padding: 2px; display: inline-block;">0</div></div> <div data-bbox="182 255 515 281">0 composed with anything is 0</div> | <div data-bbox="1384 152 1732 321"><div data-bbox="1525 152 1671 178">[Char]×- Code</div><div data-bbox="1384 186 1732 284"><div data-bbox="1563 226 1595 247" style="border: 1px solid black; padding: 2px; display: inline-block;">0</div></div><div data-bbox="1643 292 1652 318">1</div></div> |